

DESIGN, IMPLEMENTATION, AND VERIFICATION OF AN RV32I RISC-V SOC WITH INTEGRATED 4x4 SYSTOLIC ARRAY ACCELERATOR IN SKYWATER 130nm ASIC FLOW

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1. Executive Summary

This report documents the architectural design, Register-Transfer Level (RTL) implementation, transaction-level Universal Verification Methodology (UVM) verification, and complete physical design implementation (RTL-to-GDSII) of a 32-bit RISC-V System-on-Chip (SoC) tailored for matrix multiplication acceleration. The integrated architecture incorporates a single-cycle RV32I scalar core, an on-chip 1KB synthesizable scratchpad SRAM, an address-decoding interconnect bridge, and a high-throughput 2D mesh 4x4 systolic array accelerator driven by an AXI4-Lite master-slave interface. The design features transparent hardware stalls, allowing seamless multi-cycle matrix transfers without software polling overhead.

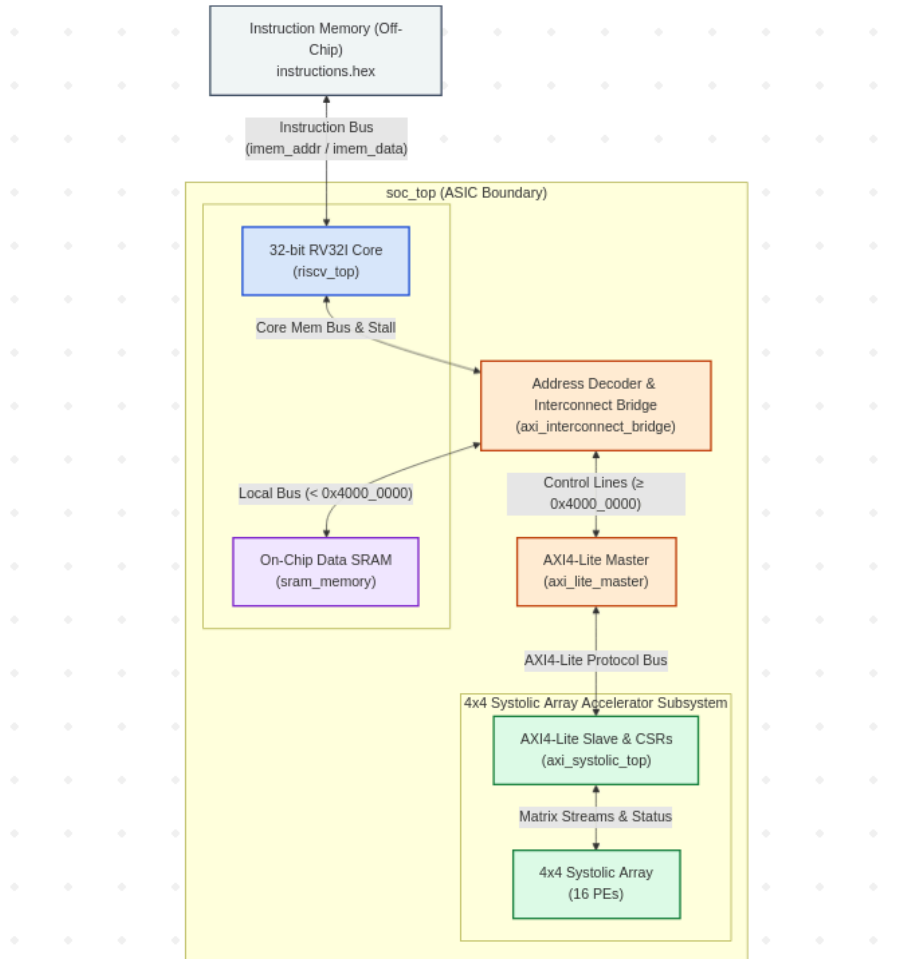
The physical realization was executed targeting the open-source SkyWater 130nm process node (sky130A) with the high-density standard cell library (sky130_fd_sc_hd). Complete physical layout closure was achieved through the LibreLane / OpenLane EDA flow across all 80 stages, delivering a clean 1:1 aspect-ratio floorplan with 43.16% core utilization across a 0.44 mm² die area.

Comprehensive functional and physical signoff confirmed:

- **Pipeline Execution:** 100% functional pass across direct assembly test suites verifying core arithmetic, memory operations, and control branches.
- **UVM Verification:** Achievement of 100.00% functional coverage and 832/832 scoreboard matches (0 errors) across constrained-random sequences and saturation corner cases (0 x 0 to 255 x 255).
- **Physical & Timing Signoff:** Total dissipation of 9.42 mW and zero setup/hold timing violations at 50 MHz (+5.21 ns WNS slack at the nominal typical corner, supporting up to 67.6 MHz), alongside fully clean DRC, LVS, and Antenna verification checks.

2. System Architecture & Interconnect Topology

The SoC comprises three primary sub-blocks: the scalar processing core, memory and interconnect subsystem, and the matrix multiplication accelerator.



2.1. 32-bit RISC-V Scalar Core (RV32I)

The CPU executes the standard integer instruction set:

- **Program Counter & State Management:** Direct generation of PC vectors with branch/jump calculation and stall capability.
- **Register File:** 32 general-purpose 32-bit dual-read single-synchronous-write registers, with register `x0` permanently hardwired to `0x00000000`.
- **Arithmetic Logic Unit (ALU):** Full RV32I arithmetic, boolean, comparison (SLT, SLTU), and multi-bit shifter logic (SLL, SRL, SRA).
- **Control & Branch Units:** Instruction opcode/funct parsing into decoded pipeline control strobes.

2.2. Address Decoding & Interconnect Bridge

Memory accesses generated by the processor are segmented by `axi_interconnect_bridge` based on address boundaries:

- **Local Data Memory Region (< 0x4000_0000):** Accesses route straight to the local 1KB synthesizable scratchpad SRAM without pipeline stalls.
- **Hardware Accelerator Region (>= 0x4000_0000):** Accesses divert to the AXI4-Lite Master interface. During multi-cycle bus transactions, the bridge automatically asserts `core_stall` to freeze the Program Counter and Register File write pipelines until the transaction finishes.

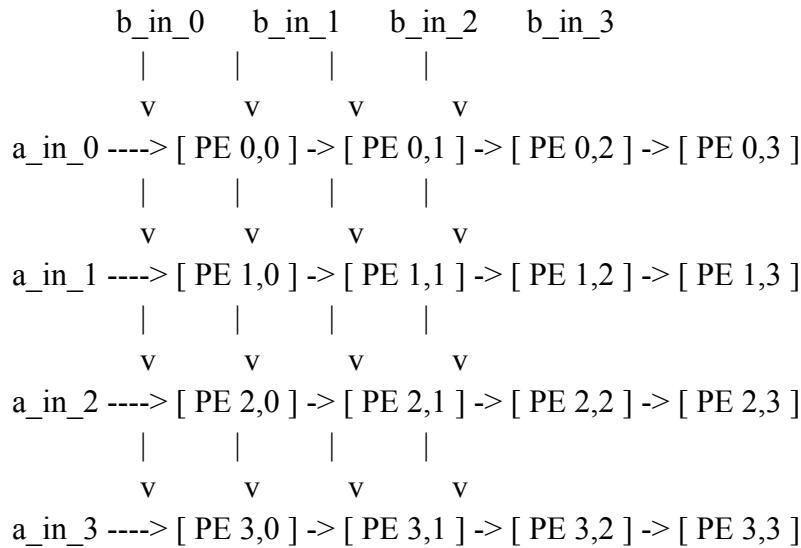
2.3. AXI4-Lite Protocol Bridge

The AXI4-Lite master engine translates core operations into standard 5-channel bus protocol handshakes:

- **Read Channels:** Address Read (`ARADDR`, `ARVALID`, `ARREADY`) and Data Read (`RDATA`, `RRESP`, `RVALID`, `RREADY`).
- **Write Channels:** Address Write (`AWADDR`, `AWVALID`, `AWREADY`), Data Write (`WDATA`, `WSTRB`, `WVALID`, `WREADY`), and Write Response (`BRESP`, `BVALID`, `BREADY`).

3. Hardware Accelerator & Systolic Architecture

The accelerator integrates an output-stationary 2D systolic grid capable of performing a 4 x 4 matrix dot-product multiplication ($C = A \times B$) in hardware.



3.1. Processing Element (PE) Design

Each individual PE incorporates:

- An 8-bit horizontal registered feedthrough ($a_{out} \leq a_{in}$).
- An 8-bit vertical registered feedthrough ($b_{out} \leq b_{in}$).
- A 32-bit accumulation register computing $C[i][j] = C[i][j] + (A[i][k] * B[k][j])$ on every clock cycle where compute is active.

3.2. Skewed Data Feed Sequencing

Correct spatial and temporal wave-front movement requires rows and columns of matrices A and B to be systematically skewed across clock steps ($k = \text{step} - 1 - \text{index}$). The execution FSM triggers this automated zero-padded feed across 16 active computation cycles.

3.3. Memory-Mapped Register Specification (CSR)

Offset Address	Register Identity	Direction	Functional Description
0x4000_0000	CTRL_START	Write	Bit [0] triggers computation sequence
0x4000_0004	STATUS	Read	Bit [0] = Busy, Bit [1] = Done
0x4000_0010 – 0x4000_001C	MAT_A_ROW[0:3]	Write	Packed 4-element row vector (8 bits each)
0x4000_0020 – 0x4000_002C	MAT_B_COL[0:3]	Write	Packed 4-element col vector (8 bits each)

0x4000_0040 – 0x4000_007C	MAT_C_RES[0:15]	Read	16 individual 32-bit dot product results (C_00 to C_33)
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4. Verification & Simulation Results

Verification was performed through directed test programs and a closed-loop UVM testbench suite.

4.1. Core Architectural Directed Tests

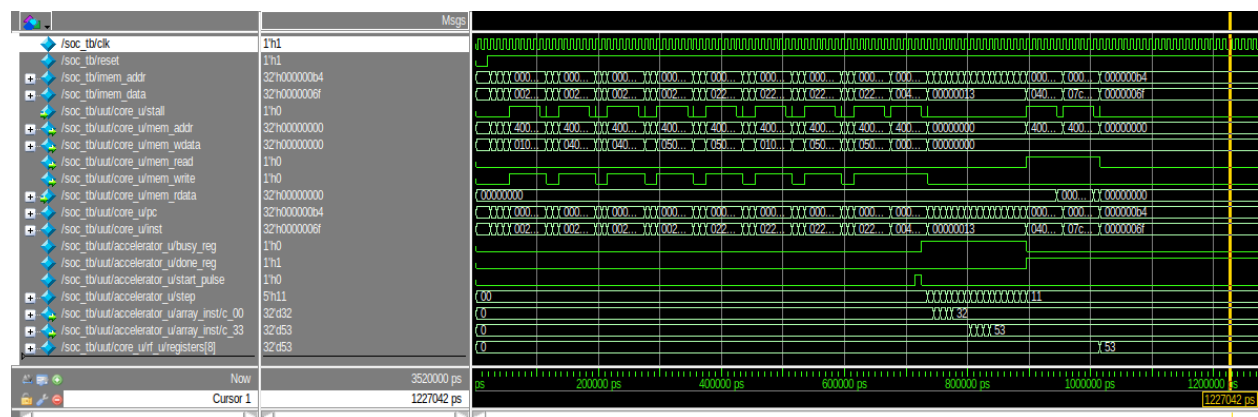
The core was verified using self-checking test programs assessing arithmetic operations, memory indexing, logical execution, sign extensions, and branch handling.

Test Target	Verilog Expression / Condition	Register Checked	Result Status
Register x0 Reset Lock	Zero protection check	rf_u.registers[0] == 0	PASS
Bitwise Boolean Group	Logical OR, XOR, AND	x4=768, x5=768, x6=0	PASS
Arithmetic Subtraction	Integer subtraction	x7 == 256	PASS
Bit Shift Sign Extensions	Arithmetic vs. logical shift	x10=0x3FFFFFFC, x11=0xFFFFFFF	PASS
Comparison Instructions	Signed vs. unsigned comparison	x12 == 1, x13 == 0	PASS

Upper Immediate Loading	LUI parsing	x14 == 0x12345000	PASS
Unconditional Jump Paths	JAL execution	x16 == 0, x17 == 1	PASS

4.2. Full SoC & Interconnect Simulation

The full system was validated by simulating `soc_tb` running matrix operations in assembly via `instructions.hex`.



```

Transcript:
# =====
# STARTING FULL SOC + 4x4 SYSTOLIC ACCELERATOR TEST
# =====
# [AXI WRITE] Time=95000 | Offset=0x10 | WDATA=0x01040101
# [AXI WRITE] Time=175000 | Offset=0x14 | WDATA=0x04040202
# [AXI WRITE] Time=255000 | Offset=0x18 | WDATA=0x04040304
# [AXI WRITE] Time=335000 | Offset=0x1c | WDATA=0x05030505
# [AXI WRITE] Time=415000 | Offset=0x20 | WDATA=0x05050205
# [AXI WRITE] Time=495000 | Offset=0x24 | WDATA=0x01040405
# [AXI WRITE] Time=575000 | Offset=0x28 | WDATA=0x05040105
# [AXI WRITE] Time=655000 | Offset=0x2c | WDATA=0x05010302
# [AXI WRITE] Time=715000 | Offset=0x00 | WDATA=0x00000001
# >>> [START PULSE TRIGGERED] Time=725000 | Computing A x B... <<<
#
# ===== MATRIX A (INPUT) =====
# [1, 1, 4, 1]
# [2, 2, 4, 4]
# [4, 3, 4, 4]
# [5, 5, 3, 5]
#
# ===== MATRIX B (INPUT) =====
# [5, 5, 5, 2]
# [2, 4, 1, 3]
# [5, 4, 4, 1]
# [5, 1, 5, 5]
#
# ===== COMPUTED MATRIX C (OUTPUT: A x B) =====
# [32, 26, 27, 14]
# [54, 38, 48, 34]
# [66, 52, 59, 41]
# [75, 62, 67, 53]
#
# --- RISC-V CORE REGISTERS ---
# x1 (Base) = 0x40000000
# x8 (C33) = 53
# =====
#
# ** Note: $finish : /home/cl4/Desktop/CAPSTONE PROJECT/RISC_V_CORE/SOC_TB/soc_tb.sv(71)
# Time: 3520 ns Iteration: 0 Instance: /soc_tb
# 1
# Break in Module soc_tb at /home/cl4/Desktop/CAPSTONE PROJECT/RISC_V_CORE/SOC_TB/soc_tb.sv line 71
VSIM 102>

```

The computation matched across all 16 results, with the final element ($C_{33} = 53$) loaded into general register `x8`.

4.3. UVM Verification Environment

A constrained-random UVM environment (`uvm-1.1d`) was implemented to stress test the accelerator subsystem over the AXI4-Lite bus interface.

1. Transaction Item (`systolic_seq_item`): Defines the pin-independent transaction data object (`uvm_sequence_item`). It encapsulates randomized operational fields including operation type (`READ / WRITE`), 32-bit addresses (`addr`), 32-bit write data (`data`), read data (`rdata`), and 2-bit bus responses (`resp`). Hard constraints (`c_addr_align`) enforce 4-byte word alignment (`addr[1:0] == 2'b00`) restricted strictly within the accelerator memory-mapped aperture (`0x00` to `0x7C`).

2. Physical Interface (`systolic_if`): Serves as the signal-level abstraction layer bridging the SystemVerilog class-based testbench to the DUT. It encapsulates the five standard AXI4-Lite channels: Write Address (`AW`), Write Data (`W`), Write Response (`B`), Read Address (`AR`), and Read Data (`R`). Dedicated directional modports (`master_mp` for Driver, `monitor_mp` for Monitor) enforce protocol compliance and prevent synthesis/simulation contention.

3. Protocol Driver (`systolic_driver`): Retrieves sequence items from the sequencer via standard TLM `seq_item_port` and unrolls high-level read/write transactions into pin-accurate AXI4-Lite handshake transitions. It asserts `AWVALID/WVALID` with simultaneous `BREADY` sampling for write channels, and asserts `ARVALID/RREADY` handshakes for read channels while observing clock transitions.

4. Bus Monitor (`systolic_monitor`): Passively captures signal transitions on the virtual interface (`vif.monitor_mp`) without driving pins. It monitors address/data handshakes across both write (`BVALID && BREADY`) and read (`RVALID && RREADY`) channels, reconstructs complete `systolic_seq_item` transaction packets, and broadcasts them via its `item_collected_port` to analysis subscribers.

5. Verification Agent (`systolic_agent`): Encapsulates the Sequencer (`uvm_sequencer`), Driver (`systolic_driver`), and Monitor (`systolic_monitor`) into a unified verification unit. In active mode (`UVM_ACTIVE`), it manages the TLM connection binding the sequencer export to the driver port (`driver.seq_item_port.connect`).

6. Golden Scoreboard (`systolic_scoreboard`): Acts as the automated check and validation model. It receives broadcast transactions via `uvm_analysis_imp`:

- Unpacks 32-bit row/column registers into internal 4x4 byte-level matrices (A at `0x10–0x1C`, B at `0x20–0x2C`).

- Computes reference golden dot products ($A \times B$) upon detection of the start trigger at offset `0x00`.
- Compares simulated readbacks from result offsets `0x40` through `0x7C` with expected values, logging match and mismatch counters.

7. Functional Coverage Subscriber (`systolic_coverage`): Subscribes to monitor transfers (`uvm_subscriber`) and samples the `cg_axi_systolic` covergroup. It measures comprehensive functional verification across four critical bins:

- `cp_op`: Read and Write bus operations.
- `cp_wr_addr`: Validated hits to Control (`0x00`), Matrix A rows (`0x10–0x1C`), and Matrix B columns (`0x20–0x2C`).
- `cp_rd_addr`: Reads targeting the Status register (`0x04`) and all 16 output registers (`0x40–0x7C`).
- `cp_data`: Data boundary distributions checking minimum zeros (`8'h00`), intermediate values (`[8'h01:8'hFE]`), and maximum saturation levels (`8'hFF`).

8. Environment Layer (`systolic_env`): Top-level UVM component container that instantiates the agent, scoreboard, and coverage subscriber. In its `connect_phase`, it establishes parallel TLM broadcasts connecting `agent.monitor.item_collected_port` simultaneously to `scoreboard.item_collected_export` and `cov.analysis_export`.

9. Test Sequences (`systolic_sequence`):

Generates targeted stimuli pipelines:

- **Corner Case 1 (All-Zeros):** Injects zero matrices (0×0) to verify reset state clearing and null arithmetic integrity.
- **Corner Case 2 (Full Saturation):** Injects maximum byte values (255×255) to stress 32-bit accumulator capacity and prevent integer overflow wrap-around.
- **Stress Test (Randomized Operations):** Executes 50 consecutive back-to-back randomized matrix multiplication iterations, tracking dynamic data variations across all 16 PEs.

10. Top-Level Test (`systolic_base_test`): Controls the overall simulation phase execution. It raises UVM objections, constructs and executes the `systolic_random_seq` sequence on the agent's sequencer, handles execution time budgets, and drops objections upon completion.

11. UVM Package Wrapper (`systolic_pkg`): A centralized SystemVerilog package compiling the entire verification IP suite in sequential order to maintain type resolution and namespace isolation.

12. Simulation Harness (tb_top): The top-level testbench module instantiating the 50 MHz clock source, power-on reset generator, physical interface `systolic_if`, and the DUT (`axi_systolic_top`). It registers the virtual interface into `uvm_config_db` and initiates the test execution cycle via `run_test("systolic_base_test")`.

```

# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 122270000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][1] = 854 (Expected: 854)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 122350000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][2] = 15738 (Expected: 15738)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 122430000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][3] = 2684 (Expected: 2684)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(70) @ 123170000: uvm_test_top.env.scoreboard [SCB] Golden Matrix Multiplied successfully!
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 123630000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[0][0] = 5535 (Expected: 5535)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 123710000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[0][1] = 15621 (Expected: 15621)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 123790000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[0][2] = 13776 (Expected: 13776)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 123870000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[0][3] = 28413 (Expected: 28413)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 123950000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[1][0] = 9225 (Expected: 9225)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124030000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[1][1] = 26035 (Expected: 26035)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124110000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[1][2] = 22960 (Expected: 22960)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124190000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[1][3] = 47355 (Expected: 47355)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124270000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[2][0] = 2925 (Expected: 2925)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124350000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[2][1] = 8255 (Expected: 8255)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124430000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[2][2] = 7280 (Expected: 7280)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124510000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[2][3] = 15015 (Expected: 15015)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124590000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][0] = 4050 (Expected: 4050)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124670000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][1] = 11430 (Expected: 11430)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124750000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][2] = 10080 (Expected: 10080)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(81) @ 124830000: uvm_test_top.env.scoreboard [SCB_PASS] MATCH: C[3][3] = 20790 (Expected: 20790)
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_test.sv(32) @ 124950000: uvm_test_top [TEST] Verification Sequence Completed.
# UVM INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 124950000: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_coverage.sv(59) @ 124950000: uvm_test_top.env.cov [COV_REPORT] Achieved Functional Coverage = 100.00 %
# UVM INFO /home/hassan/Documents/RISC_V_SYSTOLIC_ARRAY/uvm_tb/systolic_scoreboard.sv(92) @ 124950000: uvm_test_top.env.scoreboard [SCB_FINAL] Total Matches = 832 | Total Mismatches = 0
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM INFO : 895
# UVM WARNING : 0
# UVM ERROR : 0
# UVM FATAL : 0
# ** Report counts by id
# [COV_REPORT] 1
# [Questa UVM] 2
# [RNTST] 1
# [SCB] 52
# [SCB_FINAL] 1
# [SCB_PASS] 832
# [SEQ] 3
# [TEST] 2
# [TEST_DONE] 1
#
# *** Info: Finish ***

```

UVM Verification Summary

- UVM Warnings / Errors / Fatalities: 0 / 0 / 0
- Total Scoreboard Transaction Comparisons: 832
- Total Scoreboard Matches: 832 (100.00% matching accuracy)
- Total Scoreboard Mismatches: 0
- Achieved Functional Coverage: 100.00%

5. Physical Design Implementation (SkyWater 130nm)

Physical design implementation was executed using the open-source LibreLane / OpenLane 2/3 ASIC flow across all 80 stages.

5.1. Floorplan & Power Distribution Network (PDN)

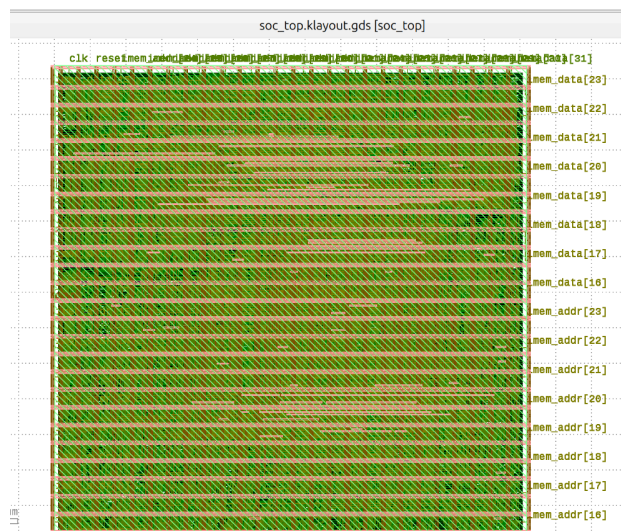
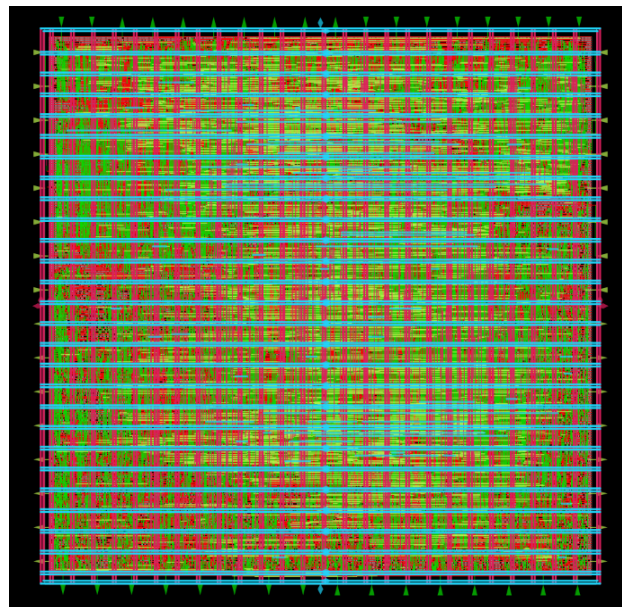
- **Floorplanning:** Implemented with relative sizing (`FP_CORE_UTIL: 40`), resulting in a 1:1 aspect ratio that shortened interconnect lengths.
- **PDN Architecture:** Uses wide supply rings on Metal 4 and Metal 5 with alternating vertical and horizontal straps (`VPITCH: 25`, `HPITCH: 25`), maintaining uniform voltage distributions.

5.2. Clock Tree Synthesis (CTS)

TritonCTS synthesized a balanced clock network using low-skew clock buffers (`sky130_fd_sc_hd_clkbuf_4` and `sky130_fd_sc_hd_clkbuf_8`). Buffer structures and skew characteristics were optimized to prevent race conditions.

5.3. Routing & Physical Verification

- **Global & Detailed Routing:** Performed with TritonRoute using DRC-aware routing parameters.
- **Antenna Protection:** 75 antenna diodes were inserted, clearing all charge-collection antenna rules.
- **Signoff Verifications:** Tapeout DRC rules (Magic & KLayout) and Layout-vs-Schematic (Netgen LVS) checks were validated with zero errors.



6. Implementation Metrics & Physical Signoff Data

6.1. Physical Area & Utilization Summary

Design Metric	Reported Value	Units / Notes
Technology Process	SkyWater 130nm (sky130A)	High Density Library (sky130_fd_sc_hd)
Die Dimensions (W × H)	658.44 × 669.16	um
Total Die Footprint Area	0.4406 (440,602 um ²)	mm ²
Core Dimensions (W × H)	647.22 × 647.36	um
Total Core Usable Area	0.4189 (418,984 um ²)	mm ²
Standard Cell Area	0.1808 (180,851 um ²)	mm ²
Core Utilization	43.16%	Dynamic relative floorplan density
Standard Cell Instances	21,372	Active logic standard cells
Sequential Elements (DFFs)	1,780	Pipelined/architectural flip-flops
Combinational Gate Elements	11,520	Inverters, AOI/OAI, arithmetic gates

Buffer / Inverter Elements	1,825	Slew and drive buffers
Filler & Tapcell Elements	77,705	71,705 Fillers + 6,000 Tapcells
Routed Wirelength	811.46	mm (811,468 um)
Total Via Placements	130,878	Single-cut vias
I/O Terminal Pin Count	68	Peripheral boundaries

6.2. Power Dissipation & IR Drop Signoff

Power Component	Analysis Metric	Value	Units
Dynamic Dissipation	Internal Cell Power	6.474	mW
	Net Switching Power	2.948	mW
Static Dissipation	Subthreshold Leakage	0.429	uW (4.29 x 10 ⁻⁷ W)
Total Core Power	Typical Signoff Power	9.423	mW
IR Drop Integrity	Worst-Case Voltage Drop	0.040	mV (4.02 x 10 ⁻⁵ V)
	Average Voltage Drop	0.0035	mV (3.55 x 10 ⁻⁶ V)

6.3. Static Timing Analysis (Multi-Corner Signoff)

Operating Corner	Process / Temp / Volt	Setup WNS (ns)	Hold WNS (ns)	Setup TNS (ns)	Hold Status
Typical (nom_tt)	25°C, 1.80V	+5.216	+0.306	0.000	MET (Clean)
Fast-Fast (nom_ff)	-40°C, 1.95V	+9.178	+0.152	0.000	MET (Clean)
Slow-Slow (nom_ss)	100°C, 1.60V	-4.665	+0.762	-1966.26	MET (Clean)

- **Signoff Target Frequency:** 50.0 MHz (T_{period} = 20.0 ns)
- **Operating Frequency (Typical Condition):**
 $F_{max} = 1 / (T_{target} - WNS_{typical})$
 $F_{max} = 1 / (20.0 \text{ ns} - 5.216 \text{ ns}) = 1 / 14.784 \text{ ns} \approx \mathbf{67.64 \text{ MHz}}$
- **Signal Integrity:** 0 Max Slew violations, 0 Max Capacitance violations, 0 Hold timing violations.

6.4. Physical Verification Checklist

Verification Domain	Signoff Tool	Unresolved Violations	Signoff Status
Detailed Routing DRC	TritonRoute Engine	0	PASSED (Clean)
Physical Geometry DRC	Magic & KLayout	0	PASSED (Clean)
Layout vs. Schematic (LVS)	Netgen Engine	0 devices / 0 nets	PASSED (Clean)
Antenna Violations	OpenROAD Antenna Rule Checker	0 nets / 0 pins	PASSED (Clean)
Power Grid Violations	OpenROAD PSM (VPWR & VGND)	0	PASSED (Clean)

7. Conclusion

This project successfully designed, implemented, and verified a 32-bit RV32I RISC-V SoC with an integrated 4x4 systolic matrix-multiplication accelerator. Using a modular bus topology, the system integrates a general-purpose processor with dedicated matrix hardware over an AXI4-Lite interconnect.

Functional correctness was confirmed through both direct assembly-level test programs and a closed-loop UVM testbench suite, achieving 100.00% functional coverage across all operating boundaries and 832 consecutive scoreboard matches without error.

The physical design flow was completed through OpenLane in SkyWater 130nm. The final GDSII layout achieved signoff with zero DRC, LVS, antenna, or hold violations, dissipating 9.42 mW at an operational frequency of 67.6 MHz under nominal conditions (0.44 mm² total die footprint).